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SaudiGovSent: A Large-Scale Arabic Dataset and Benchmark for Sentiment Analysis in Mobile Government Applications

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Abstract

The rapid expansion of mobile government (m-Government) platforms in Saudi Arabia has generated large volumes of user feedback, creating an opportunity for systematic, data-driven evaluation of public digital services. This study conducts a large-scale sentiment analysis of Arabic user reviews collected from five major Saudi m-Government applications, Absher Business, Tawakkalna, Sehhaty, Nusuk, and Najiz. A dataset comprising 84,000 reviews was constructed and classified into positive and negative sentiment categories. Five Arabic transformer-based baseline models, AraBERT, ArabicBERT, CAMeLBERT, SaudiBERT, and MARBERT, were evaluated under a unified experimental framework. Among these, SaudiBERT and MARBERT achieved the strongest performance, with MARBERT obtaining an accuracy of 91.2 percent, an F1-score of 0.858, and an AUC of 0.942. Furthermore, parameter-efficient fine-tuning using QLoRA on MARBERT preserved comparable performance (F1 = 0.854) while substantially reducing computational requirements. These findings demonstrate the feasibility of scalable sentiment analysis for evaluating and improving m-Government services.

Keywords: Arabic sentiment analysis; m-Government applications; transformer-based models; Saudi m-Government; mobile app reviews; QLoRA fine-tuning

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1. Introduction

Saudi Arabia is undergoing a rapid digital transformation under Vision 2030, characterized by the modernization of public services and enhanced citizen engagement as central policy objectives. The Kingdom has achieved near-universal internet adoption; the Communications, Space and Technology Commission reports that approximately 99 percent of individuals use the internet [1]. Similarly, the National Digital Transformation Program (NDTP) highlights that sustained investments in digital infrastructure have resulted in widespread connectivity and mobile internet speeds that are among the fastest globally [1]. Empirical evidence further indicates that mobile devices serve as the primary access medium, with nearly all users relying on smartphones for online services. Consequently, Saudi Arabia ranks among the leading countries in global smartphone usage, with 98.4 percent of individuals accessing the internet via mobile phones [2,3].

This high level of digital maturity has facilitated the rapid adoption of m-Government services. Comparative benchmarking reveals substantial progress in the United Nations E-Government Development Index, where Saudi Arabia advanced from 80th position in 2003 to 38th in 2020 [4]. In alignment with Vision 2030, the government has prioritized platforms such as Absher and Tawakkalna to streamline service delivery [4]. Academic studies consistently report high levels of user satisfaction with these platforms. For instance,

survey findings indicate strong performance in terms of usability and trust [5]. Absher, a flagship m-Government platform, exemplifies this transformation, with over 23 million authenticated users and access to more than 330 electronic government services [6]. Given the increasing societal reliance on such applications, systematic analysis of user sentiment derived from app-store reviews has become essential for evaluating service effectiveness and informing continuous quality improvement [6].

Although the application of m-Government is highly invested and technologically advanced, it is still challenged with issues like usability, complexity of authentication and service interruptions, among others. These obstacles directly affect user satisfaction and can become a restriction to digital public service effectiveness. Conventional assessment mechanisms, including surveys and professional reviews, offer minimal and slow feedback on actual user experiences. Google Play Store has user reviews, which are naturally occurring, unsolicited and high-volume feedback left by actual users. They also capture the frustrations, expectations, feature gaps and satisfaction in their own words and thus are a good source of data to understand the actual user sentiment. The reviews are particularly important in the highly dynamic digital environment of Saudi Arabia, where the constant enhancement of the system is the only factor that helps to retain trust in the digital services provided by the government.

Sentiment analysis allows opinions and emotions to be extracted automatically in large volumes. The initial studies of sentiment analysis used lexicon-based methods and simple machine learning classifiers, including support vector machines and naive Bayes models [7] that needed extensive feature engineering and domain resources to attain reasonable results. Though these techniques have been applied in fields like public service and e-Government, they tend to be ineffective in dealing with non-formal, noisy and large volumes of user-generated information such as reviews of mobile applications. Consequently, the majority of earlier studies that have been conducted to examine the m-Government adoption and service quality have employed structured surveys or small datasets [8]. The most recent advancements in deep learning, particularly transformer-based language models, have shown great improvements in sentiment classification by being able to contextualize and extend it to hundreds of thousands of reviews [9]. Such models are used to provide insights based on data in digital environments in government settings, where trends in user satisfaction, frequent complaints, and systemic service problems can be identified. Nevertheless, with such methodological innovations, there are only a small number of studies that were able to systematically evaluate sentiment among major Saudi m-Government apps through a large-scale review of the app stores, which creates a significant gap in empirical analysis on how Saudi people experience mobile public services.

In this study, large user reviews on the popular m-Government mobile applications in Saudi Arabia are gathered and preprocessed and transformer-based models are used to categorize user sentiment as positive and negative. It also examines the general trends in sentiment and common user issues in these applications to understand the actual user experience. Besides, it compares full and parameter-efficient fine-tuning methods to determine their suitability for scalable sentiment analysis. The final objective is to deliver evidence-based information that can be used to enhance the m-Government service designs and the general user experiences.

This study has the following contributions:

- Curates a large-scale Arabic dataset of user reviews collected from major Saudi m-Government mobile applications, enabling sentiment analysis in realistic and operational public-service environments.

- Comparison with transformer-based baseline models pretrained on large-scale Arabic corpora is systematically conducted, providing a curated and unified evaluation of general-purpose and dialect-aware Arabic transformers for sentiment classification in Saudi m-Government applications.
- Demonstrates reduced computational cost through parameter-efficient fine-tuning, showing that QLoRA-based adaptation of MARBERT achieves near-baseline performance comparable to full fine-tuning while substantially lowering memory and training requirements.

Overall, this study presents an in-depth and data-driven evaluation of user satisfaction with m-Government apps in Saudi Arabia. It combines insights from massive user reviews with the transformer-based analysis by generating specific actionable insights that can directly contribute to the m-Government service ecosystem improvements in the Kingdom. The results also indicate the feasibility of effective fine-tuning techniques to implement sentiment analysis models in large-scale digital government systems.

2. Literature Survey

2.1. m-Government Arabic Sentiment Analysis

User-sentiment analysis is a key method for evaluating m-Government service quality. App reviews are available at scale and provide user-generated feedback that captures real service experiences, satisfaction levels, and service failures. Some of the studies have examined Saudi m-Government and their corresponding applications through app-review datasets gathered on mobile platforms and have applied various analytical approaches. Alqaryouti et al. [7] analyzed 7345 smart government app reviews with an aspect-based lexicon and rule-based framework with an accuracy of 91.64 percent, but the method used manually constructed domain lexicons and linguistic rules, which limited scalability and customization. Hadwan et al. [8] examined 7759 reviews of Saudi governmental apps in Arabic using conventional machine learning classifiers, with 78.46 percent accuracy using KNN, and noted the Arabic language and small dataset difficulty. To overcome data scale, Hadwan et al. [10] applied the Arb-Apps Review dataset of 51,767 Google Play reviews of six Saudi m-Government healthcare applications and demonstrated an accuracy of 94.38 percent using SVM with feature engineering and SMOTE, but the architecture still relied on handcrafted features and domain-specific assumptions. In addition to application stores, Binkheder et al. [11] examined Twitter posts about Saudi m-Government health apps with an SVM word embedding model, which presented an F1-score of 0.85 but introduced bias related to the platform and representativeness. Correspondingly, Alrayani et al. [12] processed 8500 app reviews through sentiment analysis and topic modeling to determine factors of user experience, albeit on a single application. Together, these investigations indicate the methodological limitations regarding the manual design of features, the variety of datasets, and generalization, thus inspiring the implementation of deep learning-based strategies to analyze the sentiment of Saudi m-Government app reviews.

2.2. Deep Learning-Based Arabic Sentiment Analysis

Sentiment analysis using deep learning has proven to be a strong alternative to the existing methods of machine learning (ML) due to its ability to overcome most of their inherent weaknesses, especially in feature engineering, contextual perception, and scalability. In contrast to ML models, which heavily depend on human-designed features, including bag-of-words or sentiment dictionaries, deep learning models are trained to learn sequential and semantic representations on raw text. Recurrent architecture studies by Alahmary et al. [13] and Alshamari [14] proved that recurrent architectures like LSTM and Bi-LSTM are substantially more effective in capturing the long-term relationships in Saudi

dialect Arabic and social media texts, having accuracies of over 94 percent. Similarly, Iqbal et al. [15] demonstrated that LSTM-based models are generalizable in large-scale consumer review data, achieving up to 97 percent accuracy without a large manual feature design, and Almaqtari et al. [16] further improved the results by combining deep neural networks with contextual embeddings to alleviate the problem of sparsity and semantic ambiguity. Regardless of such developments, deep learning models continue to be characterized by significant weaknesses, such as high computational complexity, reliance on large amounts of labeled data, limited interpretability, and a common emphasis on binary sentiment prediction. Similarly, sequential models such as LSTM suffer due to long-range global context and parallelization efficiency. These problems reveal the need for more sophisticated architectures that can model finer contextual dependencies and enhance efficiency. This requirement led to transformer-based models like BERT, which build on self-attention mechanisms to overcome many of the remaining limitations in sentiment analysis.

2.3. Transformer-Based Arabic Sentiment Analysis

Transformer-based models address the limitations of classical machine learning and deep learning models with self-attention mechanisms to learn global contextual semantics without handcrafted features or sequential processing. Recent studies show that transformers can be used in sentiment analysis, but they are underutilized in the depth of the data and procedure. Almaqtari et al. [16] eported high accuracies of 96.92 percent, 96.68 percent and 92.93 percent against an AraBERT-based hybrid model, but they tested their model on consumer review benchmarks (Amazon, hotel, and book reviews) and not on genuine government app data. When classifying dialectal sentiment using MARBERTv2, Nwesri et al. [17] achieved an F1-score of 88 percent (training) and 79 percent (test), but their experiment used a small shared-task dataset of fewer than 1100 samples, which is not generalizable. Strong performance was shown by Aljeezani et al. [18] when using MARBERT, with an F1-score of 0.89 on the AURA dataset, but the performance decreases dramatically to 0.62 in multi-class contexts, which showcases scaling issues. Moreover, Alsalem et al. [19] achieved a fine -tuned MARBERT accuracy of 77 percent on target -dependent sentiment classification but their experiments were limited by the small size of the AT-ODTSA dataset (2790 tweets) and the expensive nature of fine-tuning. Topal et al. [20] showed that QLoRA can be as effective as full fine-tuning with significantly reduced computational demands, but they applied benchmark datasets of Twitter, rather than application-store reviews. Collectively, these limitations indicate the absence of sentiment analysis at the magnitude of the app stores of Saudi m-Government services and the need to conduct scalable transformer evaluation with viable fine-tuning paradigms.

Although there is substantial progress in Arabic sentiment analysis, but the existing literature has weaknesses that leave gaps in Saudi m-Government services. Previous m-Government-oriented research utilized small to medium datasets (e.g., 7345 reviews [7] and 7759 reviews [8]) or targeted a specific application domain, such as healthcare services [10,11], which does not allow them to reflect the range of real-world government platforms. Despite their excellent performance, the most recent transformer-based approaches consider models on benchmark consumer datasets [16], shared-task corpora with small samples [17], or Twitter-based datasets [11,19], rather than large-scale app-store reviews of functioning m-Government systems. Moreover, existing transformer studies typically involve single-model contexts [12,16] or do not provide systematic comparisons between general-purpose and dialect-aware Arabic models, leaving uncertainty about optimal model selection for public-sector NLP applications. While recent work, such as Alrayani et al. [12], has examined sentiment in individual Saudi applications, no study has systematically benchmarked multiple Arabic transformer models across five diverse Saudi

m-Government applications on a large-scale balanced dataset. Although parameter-efficient fine-tuning methods such as QLoRA have demonstrated computational advantages in Arabic NLP tasks [20], their systematic evaluation within the specific context of Arabic m-Government app review sentiment classification has received limited attention, and this work contributes an applied validation in that direction. All these limitations motivate the development of a large-scale dataset and an integrated transformer-based evaluation with efficient fine-tuning approaches, as presented in this study.

Table 1 shows the summary of existing research studies on sentiment classification of m-Government application reviews.

Table 1. Summary of Existing Research Studies on Sentiment Classification of m-Government Application Reviews.

Author(s)	Objective	Dataset	Methods	Results	Limitations
Alqaryouti et al. [7]	To perform aspect-based sentiment analysis on smart government app reviews	Smart government app reviews; 7345 reviews	Lexicon- and rule-based framework	Accuracy 91.64%	Manual lexicon construction; domain dependency; limited scalability
Hadwan et al. [8]	To analyze sentiment in Saudi governmental app reviews during COVID-19	Saudi government app reviews; 7759 reviews	ML models (KNN, SVM, NB, and DT)	Best accuracy 78.46% (KNN)	Small dataset; reliance on handcrafted features
Hadwan et al. [10]	To improve sentiment classification using large-scale app reviews	Arb-app reviews; 51,767 reviews (healthcare apps)	SVM + feature engineering + SMOTE	Accuracy 94.38%	Domain-specific (healthcare); handcrafted features
Binkheder et al. [11]	To assess public sentiment toward Saudi mGov health apps	Twitter mHealth dataset; 4719 tweets	SVM + word embeddings	F1-score 0.85	Twitter-only data
Alrayani et al. [12]	To identify UX factors in Saudi super apps	Mrsool review dataset; 8500 reviews	CAMeL sentiment, Doc2Vec, and BTM	Identified 4 UX topics; sentiment accuracy 88%	Single application focus; subjective topic labeling
Alahmary et al. [13]	To improve sentiment analysis for Saudi dialect Arabic	SDCT (Twitter), 32,063 tweets	LSTM; Bi-LSTM	Accuracy 94% (Bi-LSTM)	Twitter domain; high computational cost
Alshamari [14]	To evaluate user satisfaction in Saudi telecom services	AraCust; 20,000 tweets	LSTM, GRU, Bi-LSTM, and CNN–LSTM	Test accuracy 97.03%	Domain-specific (telecom); Twitter-only
Almaqtari et al. [16]	To enhance Arabic sentiment analysis using contextual embeddings	ACRD, HARD, and LABR; 209,000 reviews	AraBERT + CNN + BiGRU	Accuracy up to 96.92%	Consumer review benchmarks
Nwesri et al. [17]	To evaluate transformers for Arabic dialect sentiment	AHaSIS dataset; 1076 reviews	MARBERTv2, AraBERT, and QARiB	F1 88% (train); 79% (test)	Very small dataset
Aljeezani et al. [18]	To introduce Arabic app review dataset and benchmark models	AURA; 29,700 (sentiment); 2900 (classification)	MARBERT, CAMeLBERT, and RNNs	F1 0.89 (sentiment); 0.62 (multi-class)	Limited dataset size; weaker multi-class performance
Alsalem et al. [19]	To study target-dependent Arabic sentiment using LLMs	AT-ODTSA; 2790 tweets	MARBERT, AraBERT, and LLMs	Accuracy and F1 0.77	Small dataset; high fine-tuning cost

3. Methodology

The following section outlines the methodological framework that was used to examine user sentiment in Saudi m-Government application reviews. It describes the dataset construction, preprocessing procedures, and the experimental design used to train and evaluate transformer-based sentiment classification models. Figure 1 presents the overall methodological framework, detailing the stages of data collection, preprocessing, sentiment classification and evaluation.

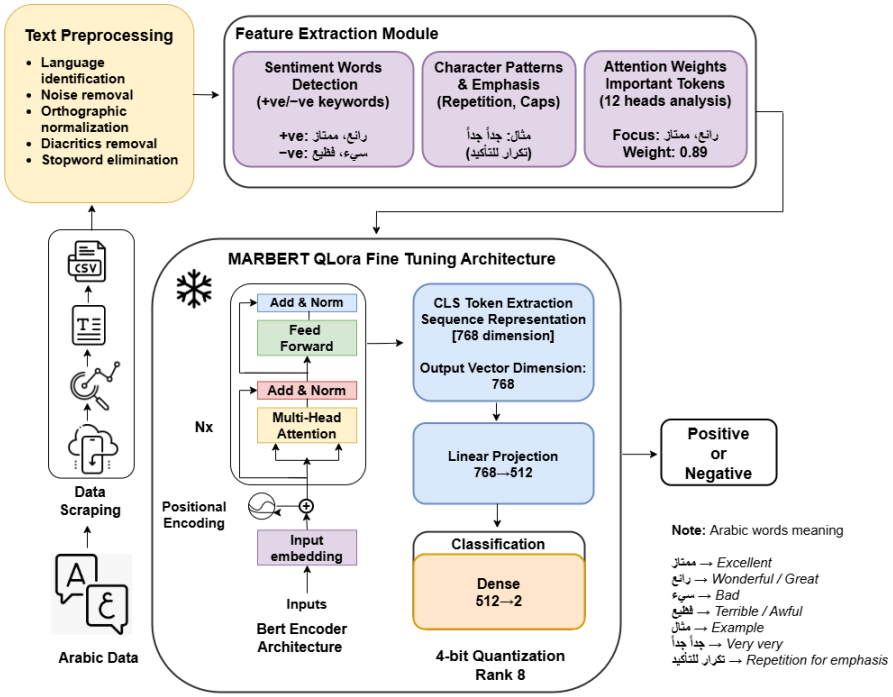


Figure 1. Methodology workflow for sentiment classification.

3.1. Dataset Description

The custom dataset was created by scraping user reviews from the Google Play Store for several Saudi government mobile applications, including Absher Business, Tawakkalna, Sehhaty, Nusuk, Najiz, Riyadh Season, RGA, and Monshaat, using Python (v3.12)-based scraping tools and the obtained data was converted into a structured CSV format. The five main applications, i.e., Absher Business, Tawakkalna, Sehhaty, Nusuk, and Najiz, were selected to allow for more detailed examination, based on their high number of reviews and the diverse roles these applications play across different areas of government.

In order to perform sentiment classification, sentiment labels were assigned based on the star ratings attached to each review. Reviews with ratings of 4 and 5 were labeled as positive, 1 and 2 were labeled as negative, and reviews assigned a rating of 3 were systematically excluded from the dataset, as such ratings typically represent neutral or ambiguous sentiments that may compromise the integrity of binary sentiment classification. The full dataset comprises approximately 212,191 Arabic user reviews, classified into two sentiment categories: 169,551 positive and 42,640 negative reviews.

To address the class imbalance between positive and negative reviews, random undersampling was applied to the majority positive class. Specifically, 42,000 reviews were randomly selected from the positive class to match the size of the negative class, which constituted 42,640 reviews in total. The balancing target of 42,000 was selected because this is the natural ceiling that is dictated by the size of the minority negative class, and thus, all negative samples are retained while minimizing information loss from the positive class. This left a balanced dataset of 84,000 reviews with equal distribution of positive

and negative sentiment. The balanced dataset was further divided into 80 percent training, 10 percent testing and 10 percent validation to give them a high statistical basis and objective analysis of sentiments of Saudi m-Government applications.

Figure 2 illustrates sample reviews, which capture the major linguistic phenomena in Arabic m-Government applications. Positive reviews are brief, while negative ones are more detailed and describe technical issues, supporting the use of dialect-aware transformer models.

Sentiment	Stars	Review (Arabic)	Review Translation
Positive	5	ممتاز ونشكر لكم حرصكم على صحة المواطنين	Excellent, and we thank you for your commitment to citizens' health
Positive	5	شيء لا يصدق من الخدمات التي عجزت عنها الكثير من الدول المتقدمة	Incredible services that many developed countries have been unable to provide
Positive	5	برنامج أكثر من رائع	An outstanding application
Negative	1	يطلب تحديث وتم التحديث لكن يرفض الدخول ويطلب التحديث مجدداً	The app requests an update; although updated, it refuses login and repeatedly asks for another update
Negative	1	لدي إشكالية، لا يصلني الإجازات المرضية من المستشفيات الحكومية	I am experiencing an issue; sick leave reports from government hospitals are not being received
Negative	2	مرات يحتاج تحديث والنت قوي ولا تفتح الصفحة	At times, it requires an update, and despite a strong internet connection, the page does not open

Figure 2. Sample dataset.

3.2. Dataset Analysis

To ensure transparency and demonstrate the quality and diversity of the collected data, this section presents a comprehensive analysis of the raw dataset comprising 212,191 Arabic user reviews scraped from the Google Play Store across five Saudi m-Government applications.

3.2.1. Per-Application Review Distribution

Selected applications' reviews distribution is presented in Table 2. The highest proportion of the dataset is Tawakkalna (126,681) and Sehhaty (73,199). Nusuk, Najiz, and Absher Business make 7936, 3833, and 542 reviews, respectively. The unequal distribution shows the different number of users and frequency of usage of these applications, with Tawakkalna and Sehhaty being flagship health and identity services, which were especially active during the COVID-19 pandemic and in its aftermath. The five were retained despite the imbalance in the applications in order to maintain the variety of government service areas, such as health, identity, legal, pilgrimage, and business services.

Table 2. Per-application review distribution in raw dataset.

Application	Total Reviews	Positive	Negative
Tawakkalna	126,681	101,931 (80.5%)	24,750 (19.5%)
Sehhaty	73,199	57,062 (78.0%)	16,137 (22.0%)
Nusuk	7936	6995 (88.1%)	941 (11.9%)
Najiz	3833	3138 (81.9%)	695 (18.1%)
Absher Business	542	425 (78.4%)	117 (21.6%)
Total	212,191	169,551 (79.9%)	42,640 (20.1%)

3.2.2. Sentiment Distribution and Class Imbalance

Table 2 indicates that the raw data has a distinctly large amount of class imbalance with 169,551 positive reviews (79.9 percent) compared to 42,640 negative reviews (20.1 percent), which results in approximately a 4:1 positive to negative ratio. The imbalance is also

present in all five applications, with the positive sentiment of 78.0 percent (Sehhaty) to 88.1 percent (Nusuk). This skew reflects the alignment between star ratings and sentiment, with positive reviews linked to higher ratings and negative reviews to lower ratings. This imbalance would cause any trained classifier to lean toward the majority positive class, thereby eliminating its ability to correctly detect negative sentiment. The majority class was randomly undersampled to create a balanced dataset of 84,000 reviews (42,000 per class) for model training and evaluation.

3.2.3. Review Length Analysis

Table 3 provides word count data of the cleaned reviews stratified by sentiment class. The mean number of words in negative reviews (9.0 words) is considerably more than the mean number of words in positive reviews (3.4 words). This pattern is typical of sentiment analysis, wherein customers who are dissatisfied are more likely to narrate their negative experiences in more depth. A majority of the reviews (77.8 percent) are short with 1 to 5 words, indicating the short and informal quality of Arabic mobile app store reviews. This highlights the importance of contextual transformer-based models that are able to obtain sentiment using limited textual hints.

Table 3. Review length statistics by sentiment class (word count).

Sentiment	Mean	Median	Std	Min	Max
Positive	3.4	1.0	5.6	1	114
Negative	9.0	5.0	10.6	1	111

Table 4 also presents the count of words per application. Nusuk reviews are the longest average length (mean: 7.2 words), whereas Absher Business reviews are the shortest (mean: 3.4 words), implying that users of more complicated services like Nusuk are more willing to give detailed reviews.

Table 4. Review length statistics by application (word count).

Application	Mean	Median	Std
Nusuk	7.2	4.0	9.6
Najiz	5.0	2.0	8.9
Sehhaty	4.9	2.0	7.9
Tawakkalna	4.1	2.0	6.6
Absher Business	3.4	1.0	5.4

3.2.4. Temporal Coverage

Table 5 contains six years of user ratings, September 2019 to December 2025. The most frequent year of COVID-19-related adoption of digital government services in Saudi Arabia has the largest amount of all reviews, 62.5 percent (132,653), with apps like Tawakkalna and Sehhaty becoming a mandatory health certification and booking device. The 2022 and subsequent reviews suggest that the m-Government services are still used even after the pandemic, which supports the relevance of these platforms.

3.2.5. Comparison with Existing Arabic Sentiment Datasets

Table 6 contextualizes the proposed dataset among the existing Arabic sentiment analysis datasets applied in m-Government and other app review studies. The proposed dataset is the largest among all comparable Arabic m-Government review datasets, comprising 212,191 raw reviews and a balanced experimental subset of 84,000 reviews. In contrast to previous datasets, which are limited to one application area at a time [12], use social

media data [11], or utilize synthetic oversampling [10], this dataset offers true multi-domain coverage of five different types of government services, directly harvested out of the Google Play Store, with a time range extending over six years (2019–2025). Moreover, the dataset uses random undersampling in class balancing, and no artificial data is made to maintain the authenticity of all the remaining samples.

Table 5. Temporal distribution of reviews by year.

Year	Reviews	Percentage
2019	168	0.1
2020	16,075	7.6
2021	132,653	62.5
2022	28,537	13.4
2023	10,365	4.9
2024	9301	4.4
2025	15,092	7.1
Total	212,191	100 percent

Table 6. Comparison of the proposed dataset with existing Arabic sentiment datasets.

Study	Size	Source	Domain	Classes
Alqaryouti et al. [7]	7345	App Store	Multi Gov	Binary
Hadwan et al. [8]	7759	Google Play	Saudi Gov	Binary
Hadwan et al. [10]	51,767	Google Play	Healthcare	Binary
Binkheder et al. [11]	4719	Twitter	Health	Binary
Alrayani et al. [12]	8500	Google Play	Single App	Binary
Aljeezani et al. [18]	29,700	App Store	Multi App	Binary/Multi
This Study	84,000	Google Play	5 Saudi Gov	Binary

3.3. Data Preprocessing

The Arabic user reviews were subjected to a systematic preprocessing pipeline in order to guarantee the linguistic consistency and analytical reliability. The pipeline was composed of language identification, noise removal and orthographic normalization. In the first stage of automated language detection, only Arabic reviews were retained, and all non-Arabic entries were excluded. Noise elimination removed English letters, numerals, punctuation, symbols, and unnecessary whitespace, while retaining Arabic script. Orthographic normalization was used to standardize Arabic variants of characters by substituting the variant forms in Alef and Hamza (آ، إ، أ) with an orthographic normal form of Alef (ا), final forms of Ya with Yea (ي) and Yea Hamza (ئ) with Ya (ي), and Ta Marbuta (ة) with Ha (ه). Moreover, diacritics and elongation marks were eliminated.

The whole preprocessing step can be mathematically formulated as follows:

$$r_i^{\text{pre}} = f_{\text{norm}}\left(f_{\text{clean}}(f_{\text{lang}}(r_i))\right) \tag{1}$$

where r_i denotes the original review text, f_{lang} represents Arabic language filtering, f_{clean} noise removal, and f_{norm} orthographic normalization. This pipeline produced clean, standardized, and linguistically coherent Arabic text, forming a robust foundation for subsequent sentiment analysis.

3.4. Transformer Models

3.4.1. AraBERT

AraBERT is implemented on the BERT-base Transformer framework with 12 encoder layers, 768-size hidden dimension, 12 heads of self-attention, and roughly 110 million parameters. It is specifically trained on large Arabic collections of data [20]. In contrast to multilingual BERT, AraBERT uses Arabic-specialized preprocessing with Farasa morphological segmentation, followed by SentencePiece subword tokenization, eliminating lexical sparsity due to prefixes and suffixes (e.g., the definite article "Al-") and leading to a large Arabic vocabulary of about 60 K subword units. AraBERT is not pretrained in this work, which is rather fine-tuned on binary sentiment classification and uses a special classification head on top of the final hidden representation of a special token known as [CLS] to achieve task-specific classification. The cross-entropy classification loss, which is defined as follows, is used to optimize the model:

$$L_{\text{cls}} = -\sum_{c=1}^{C} y_c \log \hat{y}_c \tag{2}$$

In the above equation, y_c is the ground-truth label, $\hat{y}_c$ is the predicted probability acquired through the Softmax layer and C is the total number of classes. Such a fine-tuning approach enables AraBERT to properly adjust its pretrained contextual representations to the sentiment analysis task, and thus to outperform multilingual BERT, especially on Arabic text, as it has a language-specific vocabulary and morphology-conscious design [20].

3.4.2. SaudiBERT

SaudiBERT is a BERT-base, encoder-only Transformer model with 12 encoder layers (each layer has 12 self-attention heads) and a hidden dimension of 768, pretrained on large-scale Saudi dialect text only using the Masked Language Modeling (MLM) task [21]. Contrary to the original BERT, SaudiBERT uses a SentencePiece tokenizer, implemented using the open-source library developed by Google, that contains a vocabulary of 75 k, which makes it more appropriate to informal Arabic dialectal text, emojis, and non-standard spellings, enhancing representation learning. SaudiBERT in this work is fine-tuned on binary sentiment classification by adding a task-specific linear classification head on top of [CLS] token embedding, and regularization of the architecture is performed by increasing hidden and attention dropout (0.2), L2 weight decay, learning-rate warmup and early stopping to reduce overfitting. The core self-attention process allows two-way contextualized comprehension, computing attention weights as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

where Q, K, and V denote the Query, Key, and Value matrices obtained via learned linear projections of the input representations. The query-key dot product computes token relevance scores, which are normalized using the Softmax function and applied to the value

vectors to generate contextualized representations. This mechanism enables the model to effectively capture sentiment cues that are distributed throughout the sentence. These structural and training decisions result in SaudiBERT being especially successful in sentiment analysis tasks that use noisy and dialect-filled Arabic text with high generalization.

3.4.3. ArabicBERT

ArabicBERT is trained on large-scale Arabic corpora based on the BERT-Base Transformer architecture, which contains 12 layers of encoders, 768 hidden state dimensions, and 12 self-attention heads with a total of around 110 million parameters. The model uses a WordPiece subword tokenizer as implemented in the BERT framework developed by Google, which assists with the high morphology of Arabic by breaking down words into subword pieces, and effectively reduces vocabulary sparsity as well as generalizes across the various morphs of words [22,23]. ArabicBERT is applied to binary sentiment classification in this work with the same training pipeline as demonstrated in the code, except that a task-specific linear classification head is appended after the final hidden representation of the [CLS] token, and a Softmax layer is used to estimate sentiment probabilities. The cross-entropy classification loss is used to optimize the model parameters as mentioned in Equation (2). This fine-tuning strategy enables ArabicBERT to be properly scaled to the downstream sentiment analysis task using the original Transformer architecture, producing strong results on Arabic user-generated review data.

3.4.4. MARBERT

MARBERT is built upon the BERT-Base Transformer, consisting of 12 layers of bidirectional encoders, 768-dimensional hidden states, and 12 self-attention heads, and has around 163 million parameters [24]. The model is trained on a large-scale corpus of Arabic Twitter text, both with Modern Standard Arabic and multiple dialects. In this way, good contextual representations are learned, which can be used on informal and noisy user-generated text. The text is divided into subwords using the WordPiece subword tokenizer with a big vocabulary and, therefore, is capable of capturing dialectal variation, spelling exceptions, and colloquial phrases typical of social media data. In it, MARBERT is finetuned for binary sentiment classification with the same architecture and training pipeline as SaudiBERT, with a task-specific linear classification head on top of the final hidden state of the [CLS] token, which is then followed by a Softmax layer to estimate sentiment probabilities. The loss is the cross-entropy classification loss, which is used to optimize the model parameters as mentioned in Equation (2). The given fine-tuning technique enables MARBERT to tune its pretrained dialect-aware representations to the downstream sentiment analysis task without modifying its underlying Transformer architecture, resulting in better performance than models that are trained to operate on Modern Standard Arabic.

3.4.5. MARBERT with Parameter-Efficient Tuning Using QLoRA

To enhance efficiency and generalization performance, Quantized Low-Rank Adaptation (QLoRA) is used to provide parameter-efficient fine-tuning of MARBERT under strict memory constraints. Rather than transforming the entire parameter space of the pretrained MARBERT model W_0, the base weights are fixed and quantized to 4-bit fixed point, with task-specific variations being learned using low-rank matrices. Formally, each weight matrix $W \in \mathbb{R}^{d \times k}$ is decomposed as follows:

$$W = W_0 + \Delta W, \quad \Delta W = BA, \quad A \in \mathbb{R}^{r \times k}, B \in \mathbb{R}^{d \times r}, r \ll \min(d, k) \tag{4}$$

During training, the low-rank decomposition is defined as follows: $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$, where d and k denote the input and output dimensions of the original weight matrix $W \in \mathbb{R}^{d \times k}$, respectively, and r represents the rank of the decomposition with $r \ll \min(d, k)$. Only the low-rank matrices A and B are updated, while W_0 remains fixed and stored in 4-bit quantized form. This significantly reduces the number of trainable parameters and GPU memory usage. For QLoRA configuration, the rank was set to $r = 8$, scaling factor $\alpha = 16$, and a dropout rate of 0.05 was applied to the LoRA adapters. The adapters were applied to the query and value projection matrices of each transformer layer. The model is optimized using a cosine learning rate scheduler with warm-up, label smoothing, and early stopping, while classification is performed via a softmax layer.

$$\hat{y} = \arg\max(\text{softmax}(f_\theta(x))) \tag{5}$$

where f_θ denotes the MARBERT encoder augmented with QLoRA adapters. This parametric tuning strategy enables efficient large-scale Arabic sentiment classification while preserving the expressive power of the pretrained language model.

The proposed algorithm is illustrated in Algorithm 1.

3.5. Evaluation Metrics

Four standard evaluation measures were used to evaluate the performance of the sentiment analysis models and these measures include Accuracy, Precision, Recall, and F1-score. The accuracy is the measure of the percentage of correctly classified reviews relative to all the predictions and provides an approximate measure of the effectiveness of the model. Precision is used to assess the number of reviews that were predicted to have a specific sentiment and are actually correct; hence, the capability of the model to avoid false positives. Recall is used to assess how the model can accurately capture all the instances of a specified sentiment that are relevant and important, so that it does not miss out on important user opinions and has minimal false negatives. The harmonic mean of Precision and Recall is the F1-score, and is a single balanced measure that is particularly useful when both false positives and false negatives are of significance, as frequently occurs in sentiment analysis of public service applications.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{7}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{8}$$

$$\text{F1-score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \tag{9}$$

TP represents the number of reviews that have been correctly classified as positive sentiment, TN represents the number of reviews that have been correctly classified as negative sentiment, FP represents the number of positive reviews that were incorrectly classified as negative, and FN denotes the number of negative reviews that were incorrectly classified as positive, calculated over the binary sentiment classification task on reviews of Saudi m-Government applications.

Algorithm 1 QLoRA Adapter Fine-Tuning for Arabic Sentiment Classification

Require: $\mathcal{D}$: dataset of Arabic reviews with ratings; E: number of epochs; α: learning rate; B: batch size
Ensure: Trained LoRA adapters $\mathcal{A}$ and evaluation metrics
Data Preprocessing
Load dataset $\mathcal{D}$
Remove invalid samples
Text Normalization
for each $r \in \mathcal{D}$ **do**
    $r \leftarrow \texttt{NormalizeArabic}(r)$ ▷ normalize Alef variants, Ya, Ta Marbuta; remove diacritics and Tatweel
end for
Label Assignment
for each $(r, rating) \in \mathcal{D}$ **do**
    if $rating \geq 4$ **then**
        $label(r) \leftarrow 1$
    else if $rating \leq 2$ **then**
        $label(r) \leftarrow 0$
    else
        Remove r from $\mathcal{D}$
    end if
end for
Balancing
$\mathcal{D} \leftarrow \texttt{Balance}(\mathcal{D})$ ▷ random undersampling to 42,000 samples per class
Tokenization
for each $r \in \mathcal{D}$ **do**
    $tokens(r) \leftarrow \mathcal{T}(r, \text{max_length} = 128)$
end for
Split
$(\mathcal{D}_{train}, \mathcal{D}_{val}, \mathcal{D}_{test}) \leftarrow \texttt{Split}(\mathcal{D})$
Model Setup
Load MARBERT with 4-bit quantization
Insert LoRA adapters $\mathcal{A}$
Freeze base model
Training
$F1_{best} \leftarrow -\infty$, patience_counter $\leftarrow 0$
for epoch $= 1$ to E **do**
    for each batch **B** **do**
        $\hat{y} \leftarrow \mathcal{M}(\mathbf{B})$
        $\mathcal{L} \leftarrow \texttt{CrossEntropyLoss}(\hat{y}, y)$
        Update $\theta_{\mathcal{A}}$ using AdamW optimizer with learning rate α
    end for
    Compute Accuracy, Precision, Recall, and F1-score on validation set
    if $F1_{val} > F1_{best}$ **then**
        Save best model
        $F1_{best} \leftarrow F1_{val}$
        patience_counter $\leftarrow 0$
    else
        patience_counter $\leftarrow$ patience_counter $+1$
    end if
    if patience_counter $\geq$ patience **then**
        break
    end if
end for
Test and Save Results

4. Results and Discussion

4.1. Experimental Setup

All experiments were conducted on a local workstation to provide optimal performance and efficient use of computational resources. Training and evaluation were performed using dual NVIDIA GeForce GTX 1080Ti GPUs (11 GB each), which were

used to handle the large-scale experiments effectively. Experiments were conducted in Python (v3.12) based on the PyTorch (v1.12.1) deep learning framework and HuggingFace Transformers (4.21.0) library. For the baseline experiments and comparative analysis on the data, the following models were utilized: AraBERT (aubmindlab/bert-base-arabertv02), ArabicBERT (asafaya/bert-base-arabic), CAMeLBERT (CAMeL-Lab/bert-base-arabic-camelbert-mix), and SaudiBERT (malmarjeh/bert-base-arabic-saudi). This ensures reproducibility and clarity of the experimental setup.

Table 7 highlights the key training hyperparameters and evaluation settings used throughout all experiments to ensure consistency, reproducibility, and comparability across the evaluated approaches.

Table 7. Experimental configuration and hyperparameters.

Parameter	Value
Operating system	Windows 10
Framework	PyTorch; Hugging Face Transformers
Programming language	Python
RAM	16 GB
GPU	NVIDIA GeForce GTX 1080Ti $\times$2 (11 GB each)
Storage	2 TB HDD + 1 TB SSD
Batch size	16
Epochs	10
Maximum sequence length	128
Learning rate	1×10^{-5}
Optimizer	AdamW
Loss function	Cross-entropy loss
Early stopping	Enabled

4.2. AraBERT Results

AraBERT achieved F1-scores of 93 percent negative sentiment and 85 percent positive sentiment. As indicated in the Figure 3 confusion matrix, 0.930 negative and 0.853 positive samples were accurately classified, whereas 0.147 positive samples were wrongly classified as negative and 0.070 negative samples were wrongly classified as positive, which indicates rather equal distributions of errors in comparison with the other models. The loss curves show the steep initial drop off and a subsequent stabilization at about 400 training steps, and steady alignment until about 1200 steps. The ROC curve shown in Figure 3 registers an AUC of 0.940, which means good separation between sentiment categories. The generally balanced behavior of the model is further reinforced by macro-average F1-scores of 89 percent and weighted average F1-scores of 91 percent respectively, albeit with negative sentiment classification being more forceful.

4.3. CAMeLBERT Results

During the early training, CAMeLBERT showed a fast convergence rate, and the training and validation loss levels off at around 500 training steps and consistently trained throughout. The model attained 91 percent accuracy and F1-scores of 93 percent and 85 percent in negative and positive sentiments, respectively. The confusion matrix in Figure 3 indicates that 2833 of the negative and 1148 of the positive instances were classified correctly, with a misclassification of 227 of the positive and 171 of the negative instances, respectively, which reflects a minimal bias toward negative prediction. Figure 4 reports the AUC of 0.940 with the ROC curve, which shows sufficient discriminative capability. CAMeLBERT has a stable and consistent performance in terms of sentiment categories.

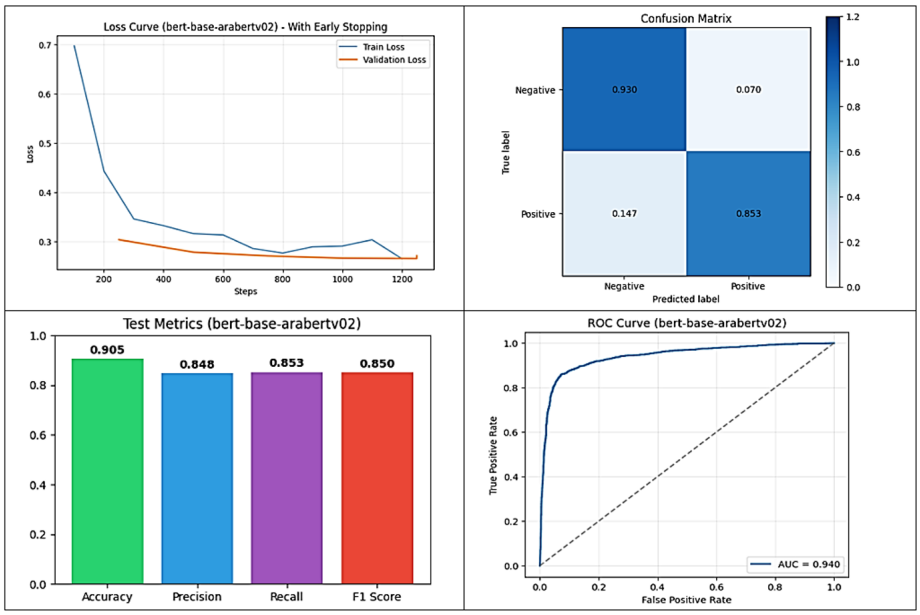


Figure 3. Evaluation results of AraBERT for sentiment classification.

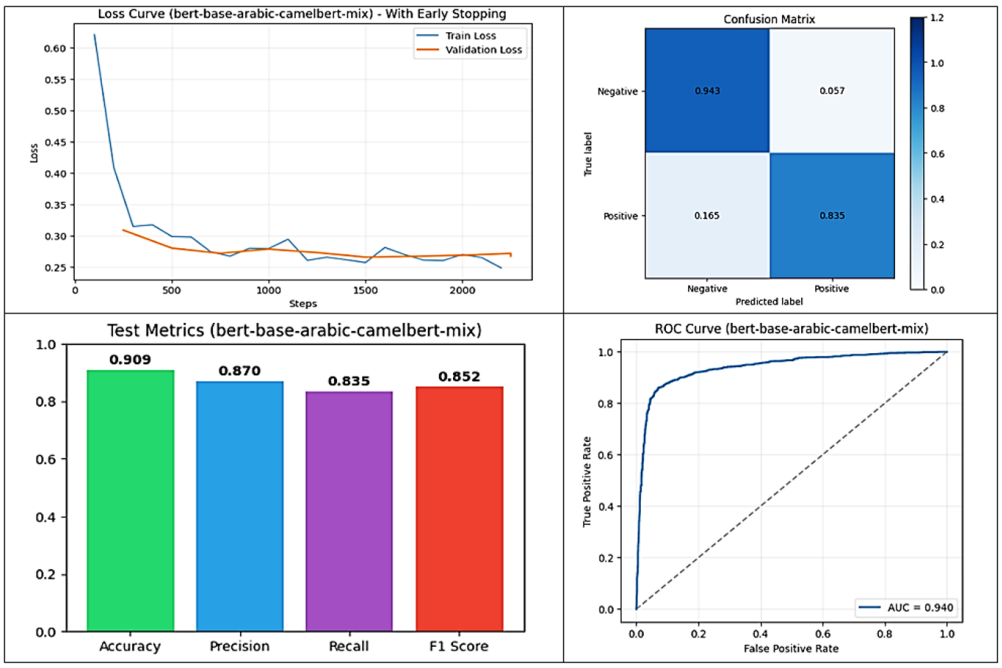


Figure 4. Evaluation results of CAMeLBERT for sentiment classification.

4.4. SaudiBERT Results

SaudiBERT attained F1-scores of 94.0 percent for the negative sentiment and 86.0 percent for the positive sentiment. As shown by the confusion matrix in Figure 5, 0.948 negative and 0.838 positive samples were identified accurately, and 0.161 positive samples were classified as negative and 0.052 negative samples as positive. The loss curves show that efficient convergence is observed at the beginning of training, with other behavioral patterns being observed afterward, and this is an important indication that the model is an effective discriminant of sentiment classes. The AUC of the ROC curve is 0.943, which is another important result showing that the model is an effective discriminant of sentiment classes. The macro-average and weighted average F1-score of 90.0 percent and 91.0 percent, respectively, represent the balanced performance, but the model, like others, demonstrates the comparatively high ability in the negative sentiment classification.

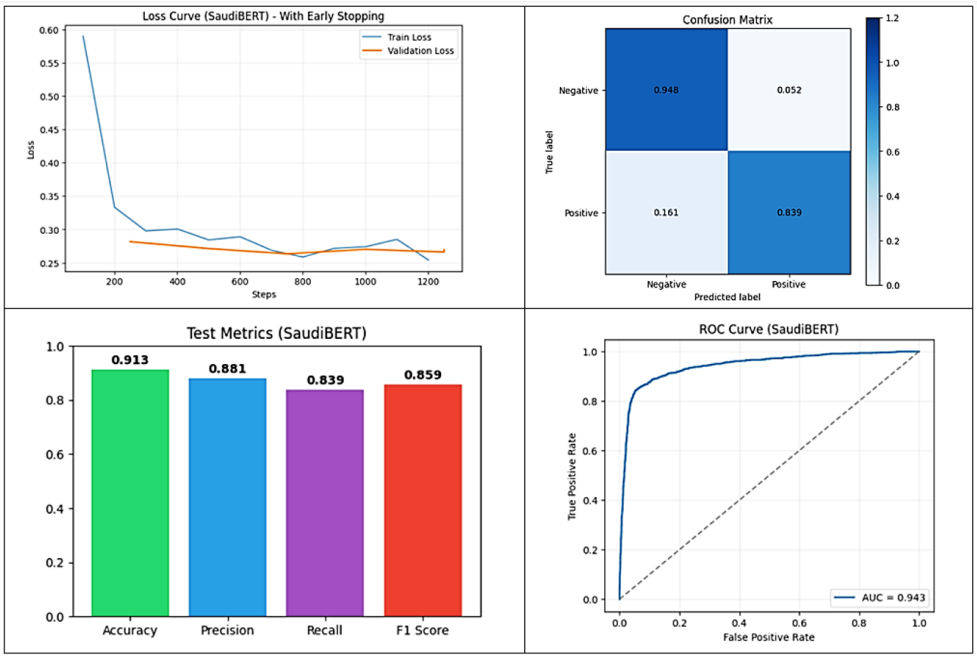


Figure 5. Evaluation results of SaudiBERT for sentiment classification.

4.5. ArabicBERT Results

The experimental findings have shown that ArabicBERT has a good performance in binary sentiment classification of Saudi m-Government application reviews with equal positive and negative classes. This model has a steady convergence with early stopping because the training and validation curve of the loss is close, as depicted in Figure 6, that is, it is generalizing positively. The precision, recall, and F1-score of ArabicBERT on the test set are 86.1 percent, 83.5 percent and 84.8 percent, respectively, which suggests that there is a good balance between false positives and false negatives. The confusion matrix validates the large share of the correct classification in both the negative and positive classes, 0.938 and 0.835, respectively, and a fairly low error of misclassification. In addition, the ROC analysis yields an AUC of 0.941, which indicates the high extent of discrimination of the model and the effectiveness of the model in the determination of sentiment polarities in Arabic user-generated information. On the whole, these findings confirm the relevance of ArabicBERT in the sentiment analysis in Saudi m-Government application review mining.

4.6. MARBERT Results

The overall accuracy of MARBERT on the Arabic sentiment classification task was 91.2 percent. The model achieved F1 scores of 94 percent and 86 percent on the negative and the positive sentiment predictors, respectively, which shows there was a distinct difference in performance between the two classes. The confusion in Figure 7 reveals that 0.944 negative and 0.843 positive were well classified, and 0.157 positive and 0.056 negative were misclassified, respectively. This distribution suggests comparatively greater difficulty in identifying positive sentiment. The training and validation loss curves show smooth convergence early on and remain stable throughout the training process, demonstrating successful optimization without overfitting. The ROC curve has an AUC of 0.942, which indicates a powerful discriminative capacity over the classification thresholds. MARBERT has a fairly stable performance in terms of sentiment categories with a macro-average and weighted-average F1-score of 90 percent and 91 percent, respectively.

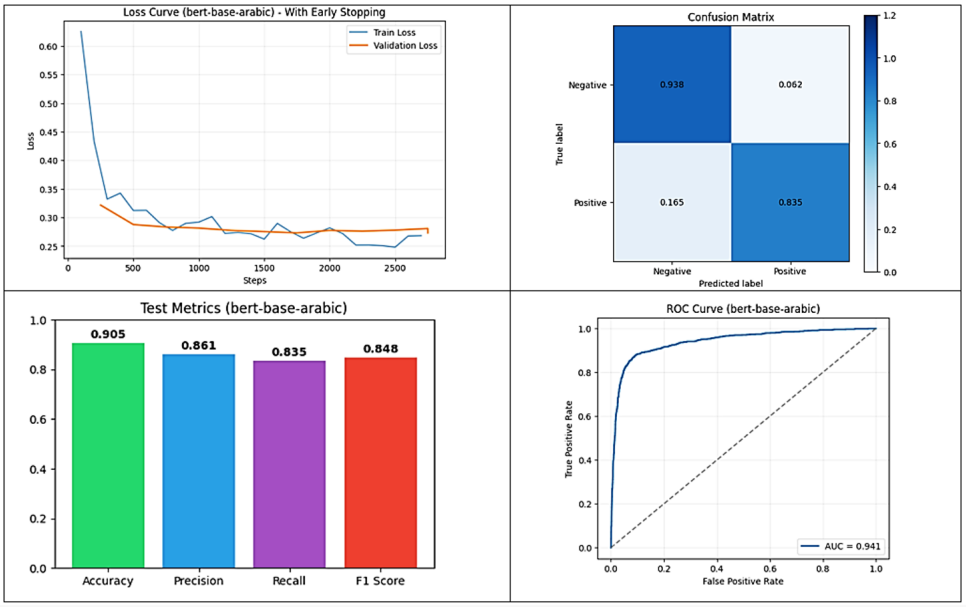


Figure 6. Evaluation results of ArabicBERT for sentiment classification.

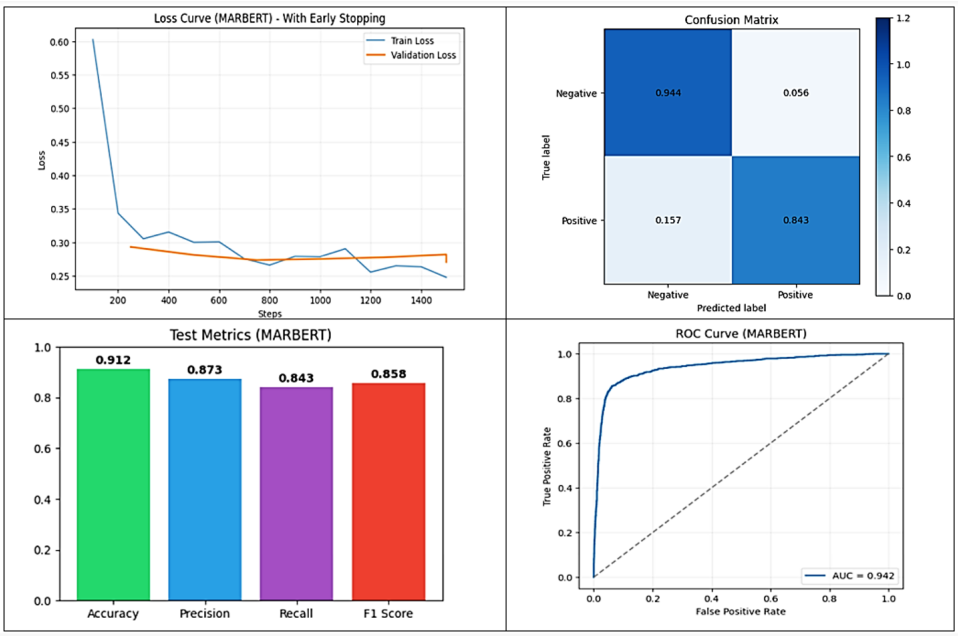


Figure 7. Evaluation results of MARBERT for sentiment classification.

4.7. MARBERT with QLoRA Fine-Tuning Results

As a quantitative measure of the effectiveness of parameter-efficient fine-tuning, MARBERT was additionally trained with the help of the Quantized Low-Rank Adaptation (QLoRA) method. The resulting model scored an accuracy of 0.910, precision of 0.866, recall of 0.843 and an F1-score of 0.854 on the test set, as compared to an F1-score of 0.858 with full fine-tuning. It is an insignificant difference in performance of less than 0.5 percent. Both training and validation loss curves in Figure 8 show steady convergence with no evidence of overfitting and the validation F1-score is always near to the fully fine-tuned model. It is important to note that recall is retained with QLoRA fine-tuning, which proves that the model does not lose its capacity to recognize positive sentiment occurrences. These findings show that QLoRA allows MARBERT to maintain almost the same predictive performance with significantly fewer trainable parameters and fewer computational needs. Therefore,

MARBERT + QLoRa provides a scalable and efficient alternative to full fine-tuning on large-scale sentiment analysis of reviews of Arabic m-Government applications.

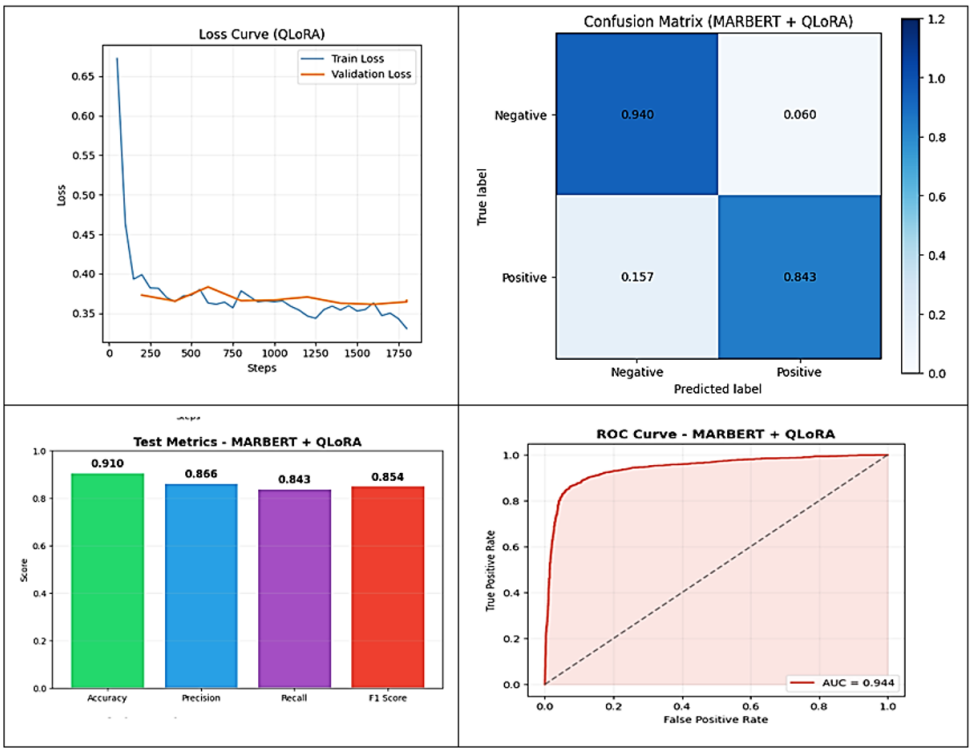


Figure 8. Evaluation results of MARBERT + QLoRA for sentiment classification.

4.8. Comparative Performance of Baseline Models

Table 8 presents a comparative summary of the five baseline transformer models evaluated in this study. Overall, all models achieved competitive performance, with accuracy values exceeding 90 percent and ROC-AUC scores consistently above 0.94, indicating promising sentiment discrimination across models. Notably, dialect-aware models exhibited marginally higher F1-scores and recall compared to general Arabic transformers, although the overall performance differences among models remained small.

Table 8. Performance comparison of sentiment classification models.

Model	Accuracy	F1-Score (Negative)	F1-Score (Positive)	Macro Average F1	AUC
SaudiBERT	91.3%	94.0%	86.0%	90.0%	0.943
MARBERT	91.2%	94.0%	86.0%	90.0%	0.942
ArabicBERT	90.5%	93.0%	85.0%	89.0%	0.941
AraBERT	90.5%	93.0%	85.0%	89.0%	0.940
CAMeLBERT	90.9%	93.0%	85.0%	89.0%	0.940
MARBERT + QLoRA	91.0%	93.0%	85.0%	89.0%	0.944

Among the evaluated baselines, SaudiBERT and MARBERT achieved the highest overall performance, obtaining the strongest F1-scores and exhibiting a balanced trade-off between precision and recall. SaudiBERT demonstrated marginally higher precision, whereas MARBERT achieved slightly higher recall, reflecting differences in error distribution observed in their respective confusion matrices. The differences are not very substantial, but the overall performance gaps are less than 0.5 percent.

General Arabic models, such as, AraBERT and ArabicBERT showed competitive performance, but they consistently performed worse than dialect-aware models, especially in recall and F1-score. CAMeLBERT showed steady performance and high separability of the classes, but had lower recall than SaudiBERT and MARBERT. Generally, the comparative findings suggest that all of the baseline models are useful in the Arabic sentiment classification, but models trained on dialect-rich corpora offer slight but consistent improvements over general Arabic transformers on real-world m-Government application reviews.

Although the overall performance is high, in all the assessed models, a similar trend of higher error rates for positive sentiment classification than for negative sentiment is shown in the confusion matrices. This unequal representation may be explained by several linguistic peculiarities of the Arabic m-Government app reviews. First, many positive reviews are extremely short statements that provide few contextual cues to the model. Second, switching between Modern Standard Arabic and the Saudi dialect creates lexical variation which cannot be well represented even by dialect-aware models like MARBERT and SaudiBERT [24]. Third, sarcastic or ironic reviews, which are one of the challenges in Arabic sentiment analysis, can contain surface-level positive words but convey negative sentiment, which can lead to misclassification. These linguistic aspects emphasize the shortcomings of binary sentiment classification on informal Arabic user-generated content and encourage future research on aspect-based and fine-grained sentiment analysis.

Despite the strong performance of the evaluated models, these results should be interpreted as baseline benchmarks rather than upper performance limits. The relatively small performance differences and the challenges identified in short and ambiguous reviews indicate that there remains significant scope for future research to improve sentiment classification on this dataset.

5. Conclusions

This work presents a large-scale sentiment analysis of Arabic user reviews of five major Saudi m-Government apps using an integrated transformer-based experimental platform. The dataset of 84,000 reviews was balanced based on 212,191 raw Arabic reviews collected in the Google Play Store over six years (2019–2025) and five different domains of government services. There were five Arabic transformer models that were assessed methodically, given the same training conditions. Dialect-sensitive models achieved higher success than their general Arabic transformers, which reported their accuracy at 91.3 percent (SaudiBERT, AUC = 0.943) and 91.2 percent (MARBERT, F1 = 0.858, AUC = 0.942). The viability of parameter-efficient fine-tuning of large-scale Arabic public sector sentiment analysis was also demonstrated by QLoRa fine-tuning of MARBERT with approximately the same performance (F1 = 0.854) but using significantly fewer computational resources. The results are added to the multi-domain Arabic m-Government benchmark and controlled comparison of the transformers between general and dialect-sensitive transformers and practical validation of QLoRA to revisit Arabic m-Government classification as a parameter-efficient fine-tuning study.

6. Limitation & Future Direction

Despite the strong baseline performance of the proposed framework on five Saudi m-Government applications, several limitations remain. First, the dataset comprises a large percentage of short and informal reviews, which can restrict the capability of models to capture more contextual and semantic details. Moreover, the fact that there are dialectal variations, sarcasm and implicit expressions of sentiment also complicates the task. The evaluation is also based on a fixed data split, and alternative validation strategies such

as cross-validation or different train-test splits may provide further insights into model generalization and robustness.

These limitations indicate that sentiment classification on this dataset is not a solved problem and remains open for improvement. Future research can consider aspect-based sentiment analysis to assess particular dimensions of service like usability, complexity, and reliability. Additionally, the analysis can be extended beyond binary classification to include neutral or mixed sentiments, providing more detailed user feedback. The relevance of the proposed benchmark can also be evaluated in other Arabic m-Government applications and the rest of the Arabic-speaking regions. Another direction that is important is to expand the dataset to larger and more varied samples across platforms and analyze variations in user feedback over time. Lastly, k-fold cross-validation can be introduced into future benchmarking studies to improve the statistical reliability of model analysis.

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Institutional Review Board Statement: The research followed the Declaration of Helsinki (1975, revised in 2013) [25]. This study did not have any ethical review or approval since the study used publicly available anonymized data obtained through the reviews of the Saudi mobile government applications on the Google Play Store, and did not involve direct contact with human subjects.

Informed Consent Statement: The consent was exempted since the research was conducted using anonymized data that is publicly available and involved no direct human interaction.

Data Availability Statement: In this study, the data were obtained from the publicly accessible reviews of the Saudi mobile government apps on Google Play Store. The data is available upon request by the respective author.

Conflicts of Interest: The author declares no conflicts of interest.

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